

Real Electronics, Learnt Virtually



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Imagine you could go to a real school, college or other learning centre to study electronics; what facilities would you get to achieve your learning target? Lectures, course materials, well-equipped labs, a competitive peer group and opportunities like workshops, seminars and contests. It might delight you to know that all these are available online these days. Whether you wish to learn the basics or top-up your skills in specific areas, it is possible to do so from the comfort of your own home—if only you have the time and dedication.

Deciding what you want

Once you have decided on learning something new, basic or advanced, you will realise upon your first Web search that there are too many opportunities waiting for you out there. You can opt for a structured online course from a reputed university and work

your way through it, or choose random resources from here and there to learn and practice what you learnt. You will find free as well as paid resources online.

What is important is that you first join a forum of like-minded people with similar learning goals and spot resources recommended by users. Make sure to check out each resource to decide if it is worth your time. If you are going to pay for something—courseware, tool or lab—make sure to read up a substantial number of reviews before signing up for it.

Your chance to sit in a classroom at any age

For free online courses, you need to look no further than edX, a non-profit initiative by Harvard and MIT. A reliable source of interactive online classes and massive open online courses (MOOCs) from the

most reputed universities in the world, edX is not just for amateurs and students but also for experts who have always wished for a chance to experience the Ivy League. On edX, you will find courses from MITx, HarvardX, BerkeleyX, UTx and many other reputed organisations.

While you can find courses on almost all subjects, readers of this magazine might be especially interested in a 16-week free course on circuits and electronics (6.002x) taught by Prof. Anant Agarwal, president of edX and professor of electrical engineering and computer science at MIT, along with other eminent faculty. The course introduces engineering in the context of lumped circuit abstraction.

Most of the top universities also have their own online

